```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import r2_score, mean_absolute_error
from sklearn.metrics.pairwise import cosine_similarity
```

```
ratings = pd.read_csv("ratings.csv")
movies = pd.read_csv("movies.csv")

print(ratings.head())
print(movies.head())
```

```
   userId  movieId  rating    timestamp
0       1        1     4.0  964982703.0
1       1        3     4.0  964981247.0
2       1        6     4.0  964982224.0
3       1       47     5.0  964983815.0
4       1       50     5.0  964982931.0
   movieId                               title  \
0        1                    Toy Story (1995)
1        2                      Jumanji (1995)
2        3             Grumpier Old Men (1995)
3        4            Waiting to Exhale (1995)
4        5  Father of the Bride Part II (1995)

                                        genres
0  Adventure|Animation|Children|Comedy|Fantasy
1                  Adventure|Children|Fantasy
2                             Comedy|Romance
3                       Comedy|Drama|Romance
4                                      Comedy
```

```
df = pd.merge(ratings, movies, on="movieId")

print(df.head())
```

```
   userId  movieId  rating    timestamp                        title  \
0       1        1     4.0  964982703.0                Toy Story (1995)
1       1        3     4.0  964981247.0        Grumpier Old Men (1995)
2       1        6     4.0  964982224.0                       Heat (1995)
3       1       47     5.0  964983815.0  Seven (a.k.a. Se7en) (1995)
4       1       50     5.0  964982931.0   Usual Suspects, The (1995)

                                        genres
0  Adventure|Animation|Children|Comedy|Fantasy
1                             Comedy|Romance
2                       Action|Crime|Thriller
3                            Mystery|Thriller
4                      Crime|Mystery|Thriller
```

```
df.drop_duplicates(inplace=True)

df['timestamp'] = pd.to_datetime(df['timestamp'], unit='s')

df.rename(columns={'genres': 'genre'}, inplace=True)
```

```
movie_views = df.groupby('movieId')['rating'].count()
df['views'] = df['movieId'].map(movie_views)
```

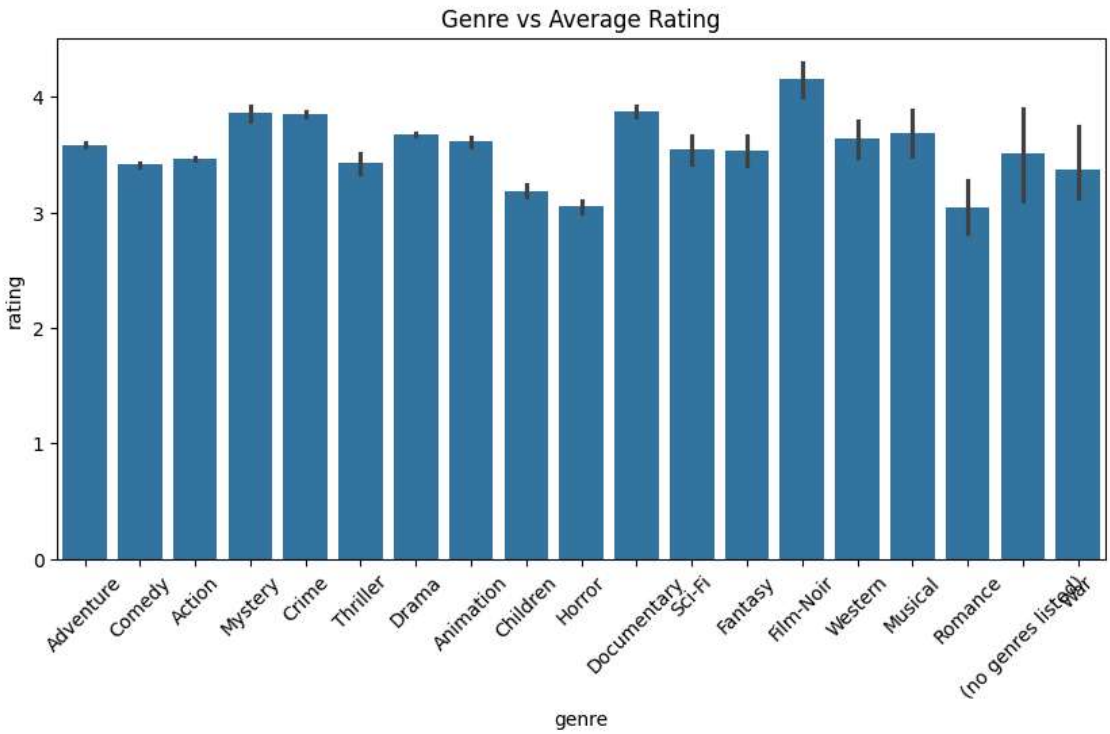
```
movie_likes = df.groupby('movieId')['rating'].mean()
df['likes'] = df['movieId'].map(movie_likes) * 100
```

```
df['popularity_score'] = (0.4 * df['views'] + 0.3 * df['likes'] + 0.3 * df['rating'])
```

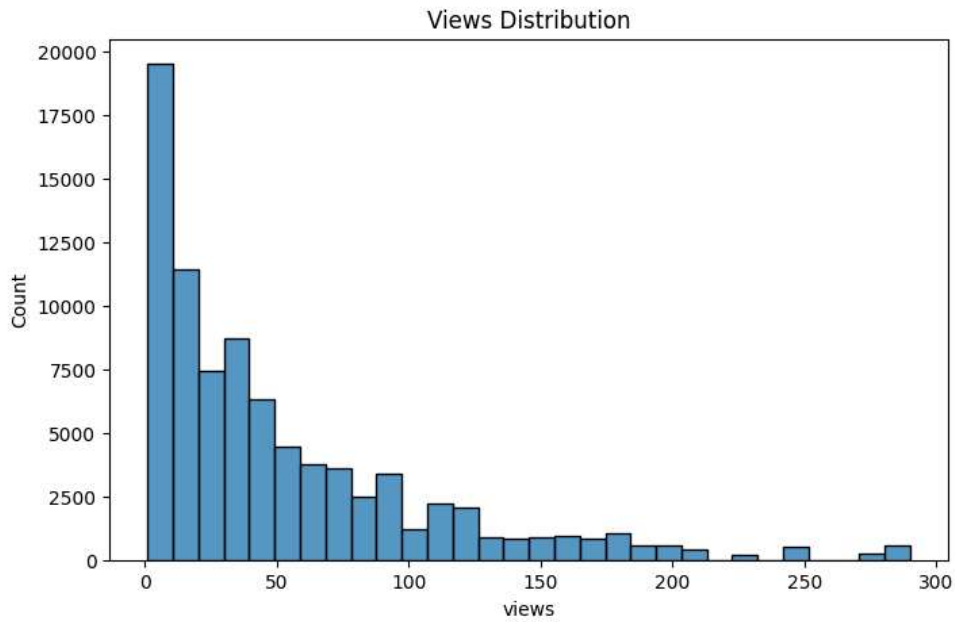
```
df['month'] = df['timestamp'].dt.month
```

```
df['genre'] = df['genre'].apply(lambda x: x.split('|')[0])

plt.figure(figsize=(10,5))
sns.barplot(x='genre', y='rating', data=df)
plt.xticks(rotation=45)
plt.title("Genre vs Average Rating")
plt.show()
```

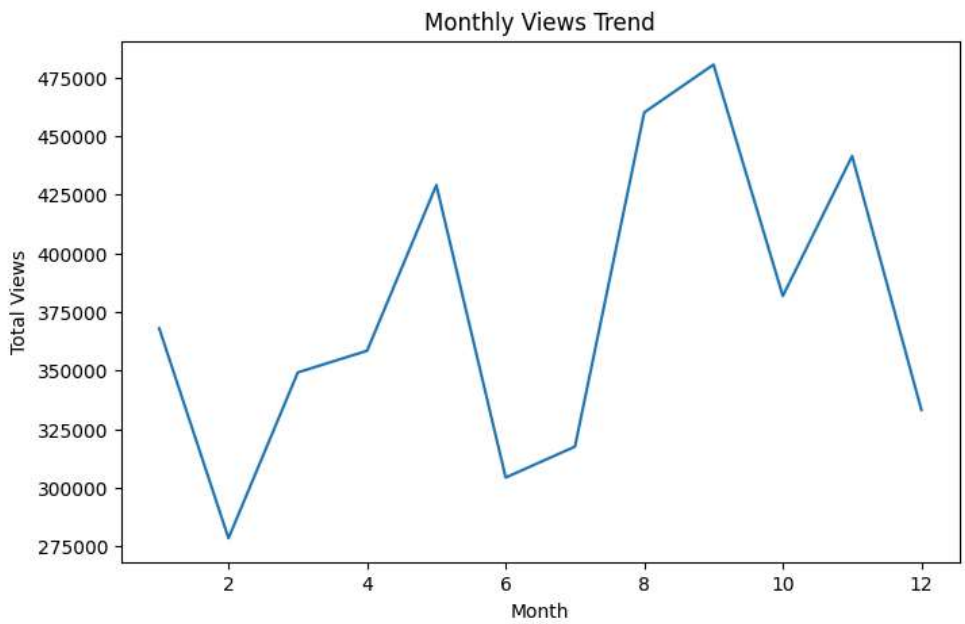



```
plt.figure(figsize=(8,5))
sns.histplot(df['views'], bins=30)
plt.title("Views Distribution")
plt.show()
```




```
monthly_views = df.groupby('month')['views'].sum()

monthly_views.plot(figsize=(8,5))
plt.title("Monthly Views Trend")
plt.xlabel("Month")
plt.ylabel("Total Views")
plt.show()
```




```
X = df[['views','likes','rating']]
y = df['popularity_score']
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

```
model = RandomForestRegressor(n_estimators=100, random_state=42)
model.fit(X_train, y_train)
```

```
RandomForestRegressor
```

```
df = df.replace([np.inf, -np.inf], np.nan)
df = df.dropna()

X = df[['views','likes','rating']]
y = df['popularity_score']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

model = RandomForestRegressor(n_estimators=100, random_state=42)
model.fit(X_train, y_train)

pred = model.predict(X_test)

print("R2 Score:", r2_score(y_test, pred))
print("MAE:", mean_absolute_error(y_test, pred))
```

```
user_movie_matrix = df.pivot_table(
    index='userId',
    columns='movieId',
    values='rating'
).fillna(0)
```

```
similarity = cosine_similarity(user_movie_matrix)

similarity_df = pd.DataFrame(
    similarity,
    index=user_movie_matrix.index,
    columns=user_movie_matrix.index
)
```

```
def recommend_movies(user_id, top_n=5):
    similar_users = similarity_df[user_id].sort_values(ascending=False)[1:6]
    similar_user_ids = similar_users.index

    recommended = user_movie_matrix.loc[similar_user_ids].mean().sort_values(ascending=False)

    return recommended.head(top_n)

print(recommend_movies(1))
```

```
df.to_csv("final_entertainment_dataset.csv", index=False)
```